

1. Practical5b.c

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if(fork()==0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls","ls","-l",NULL);
    }

    if(fork()==0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep","grep",".c",NULL);
    }
    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

```
aswika@ASWIKAINTEUR1 $ gcc Practical5b.c -o Practical5b
aswika@ASWIKAINTEUR1 $ ls -l
total 24
-rwxr-xr-x 1 root root 16224 Sep 29 05:15 Practical5b
-rw-r--r-- 1 root root 491 Sep 29 05:15 Practical5b.c
-rw-r--r-- 1 root root 0 Sep 29 05:15 a.c
-rw-r--r-- 1 root root 0 Sep 29 05:15 b.c
-rw-r--r-- 1 root root 0 Sep 29 05:15 notes.txt
-rw-r--r-- 1 root root 462 Sep 29 05:15 signal_demo.c
aswika@ASWIKAINTEUR1 $ ./Practical5b
-rwxr-xr-x 1 root root 16224 Sep 29 05:15 Practical5b
-rw-r--r-- 1 root root 491 Sep 29 05:15 Practical5b.c
-rw-r--r-- 1 root root 0 Sep 29 05:15 a.c
-rw-r--r-- 1 root root 0 Sep 29 05:15 b.c
-rw-r--r-- 1 root root 462 Sep 29 05:15 signal_demo.c
```

2. Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

```
#include <stdio.h>

#include <signal.h>

#include <unistd.h>


void handler(int sig)
{
    if(sig==SIGINT)
        printf("\nSIGINT Received\n");

    if(sig==SIGTERM)
        printf("\nSIGTERM Received\n");

    if(sig==SIGUSR1)
        printf("\nSIGUSR1 Received\n");
}


int main()
{
    signal(SIGINT,handler);
    signal(SIGTERM,handler);
    signal(SIGUSR1,handler);


    while(1)
    {
        printf("Running...\n");
        sleep(3);
    }

    return 0;
}
```

```
aswika@ASWIKAINTELLIGENCE:~$ ps -e | grep signal_demo
aswika@ASWIKAINTELLIGENCE:~$ 00:00 signal_demo
aswika@ASWIKAINTELLIGENCE:~$ $ kill -SIGUSR1 156
aswika@ASWIKAINTELLIGENCE:~$ $ kill -SIGTERM 156
aswika@ASWIKAINTELLIGENCE:~$ $ kill -SIGINT 156
aswika@ASWIKAINTELLIGENCE:~$ $ kill -9 156
```